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"Sound Transmission in Granular Media: A Review and Analysis"

Project Report

Submitted by

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1 Introduction

Sound propagation in granular materials is a fascinating and technically important phenomenon due to the complex, heterogeneous, and discrete nature of these systems. Unlike continuous solids or fluids, granular media consist of interacting macroscopic particles, whose collective mechanical behavior is governed by grain-grain contact forces and random geometric arrangements. These features make wave transmission in such materials highly nonlinear, dispersive, and attenuative. Granular materials are ubiquitous in both natural and industrial contexts—ranging from **geotechnical soils and sands to pharmaceutical powders and food grains—making** the study of their dynamic response under stress crucial for applications such as seismic wave analysis, non-destructive testing, and material characterization. In geotechnical engineering, for example, understanding how seismic waves travel through soil is essential for assessing site stability, earthquake resilience, and the potential for liquefaction. In industrial settings such as pharmaceuticals or agriculture, the mechanical response of granular media to vibration or compaction affects processing efficiency, quality control, and material handling. Despite their importance, predicting the dynamic behavior of these systems remains a challenge due to their inherent heterogeneity, contact-driven mechanics, and fluid-solid coupling in saturated conditions. This paper directly addresses this challenge by developing and validating a hybrid multiscale model that captures wave propagation and attenuation in granular beds. The classical continuum theories, like **Effective Medium Theory (EMT)**, offer a starting point by assuming affine deformation and homogeneous contact networks, predicting pressure-dependent wave speeds through **Hertz-Mindlin contact** models. However, experimental observations frequently show stronger pressure dependencies and deviations in shear response, revealing the limitations of EMT, particularly due to non-affine grain motions and evolving contact structures. Studies by Makse and colleagues, for instance, have shown that shear-induced rearrangements reduce stiffness and deviate from EMT predictions, necessitating the incorporation of microstructural heterogeneity and viscoelastic relaxation. Meanwhile, discrete element methods (DEM), originating from the work of Cundall and Strack, have provided powerful tools to simulate grain-scale interactions. Hybrid multiscale approaches that couple DEM with continuum models (e.g., poromechanics or Darcy-based fluid models) have emerged to overcome the scale gap and capture essential features like wave attenuation, liquefaction, and two-phase coupling. In this context, the present study investigates sound wave propagation and attenuation in saturated granular beds using a hybrid DEM–continuum framework. This topic was chosen because it lies at the intersection of theoretical, numerical, and applied research and offers a non-invasive method to diagnose the mechanical state of granular materials. Understanding how microstructure influences acoustic behavior not only advances granular mechanics but also has wide-reaching applications in **civil engineering, geophysics, and soft matter physics.** A major challenge in modeling sound transmission in granular systems lies in capturing the **complex interplay between individual particles and interstitial fluids**. This coupling becomes especially significant during dynamic events, such as seismic wave propagation, where changes in pore fluid pressure and soil structure can significantly affect wave speed, attenuation, and frequency content. Phenomena such as **liquefaction**, in which intergranular stresses vanish and particles behave as a suspension in two-phase flow, illustrate the need for multiphysics and multiscale modeling approaches.

Recent developments in computational modeling have brought forward hybrid frameworks that integrate discrete element methods (DEM) at the grain scale with continuum-scale models such as finite element (FEM) or finite difference (FDM) methods. In particular, approaches using DEM to simulate particle motion and Darcy's law at the macro scale for fluid flow offer

a computationally efficient yet physically grounded means of analyzing wave behavior. These hierarchical multi-scale models are capable of capturing a wide array of conditions and effects, including saturation level, wave type coupling, pressure-dependent wave speed, and post-event soil consolidation.

This report explores the fundamental physics governing sound transmission in granular systems, focusing on how grain-scale properties and interactions influence macroscopic wave behavior. We analyze experimental data, review modeling strategies, and discuss emerging hybrid numerical frameworks that integrate particle-scale and continuum-scale physics to enable realistic modeling of sound propagation in such complex media.

2 Literature Review

The study of wave propagation in granular media has evolved from classical elasticity to an interdisciplinary field incorporating micromechanics, acoustics, and statistical physics. Early studies treated granular materials as modified continua, using empirical corrections to capture grain-scale effects. Pre-1960s elastic theories, rooted in continuum mechanics, struggled with the discrete, disordered nature of these systems. In 1957, Duffy and Mindlin introduced the Effective Medium Theory (EMT), modeling granular media as homogeneous elastic solids. While EMT predicted pressure scaling of $K, \mu \propto p^{1/3}$ and $v \propto p^{1/6}$, it neglected contact disorder and assumed affine deformations.

By the 1990s, experiments had challenged these assumptions. Liu and Nagel (1992) revealed wave propagation along force chains, while Johnson et al. (1994) observed nonlinear elasticity and hysteresis under cyclic loading. These findings led to a micromechanical shift, emphasizing the role of contact geometry and localized stress transmission. In 1999, Jia et al. discovered that ultrasonic signals in granular packings exhibit both coherent low-frequency pulses and multiply scattered high-frequency codas, highlighting the coexistence of ballistic and diffusive wave transport.

To address these complexities, the 2000s saw the adoption of statistical and transport theories. Models like the Radiative Transport Equation and Bethe-Salpeter formalism, inspired by quantum field theory, were applied to wave diffusion in disordered systems. Experiments confirmed that high-frequency acoustic waves behave like photons in turbid media, undergoing multiple scattering. The Jiang-Liu model (2007) introduced nonlinear elasticity with pressure-dependent moduli, effectively capturing stress-induced anisotropy and dilatancy.

Granular materials exhibit inherently complex mechanical behavior due to disordered packing and interparticle forces. Traditional DEMs improved grain-scale modeling but often excluded angular interactions. To overcome this, recent work proposed a Kirkwood-type spring network over Delaunay tessellations, incorporating both normal and angular springs. This model enables tuning of the effective Poisson's ratio—including auxetic behavior—thereby controlling bulk and shear moduli. Such tunability is essential for manipulating sound propagation and energy dissipation.

Otsubo et al. (2021) demonstrated that softer grain inclusions increase wave attenuation via viscoelastic losses, reinforcing the importance of contact mechanics in acoustic transmission. Modern approaches now combine Discrete Element Methods (DEM), Finite Element (FEM), and Finite Difference Methods (FDM) in hybrid, multiscale frameworks to model dispersion, attenuation, and fluid-grain coupling in saturated systems. Post-2010s advancements leverage imaging (e.g., X-ray CT), machine learning, and high-resolution sensing to bridge microstructure with macroscopic wave behavior.

Today, granular materials are recognized as mesoscopic complex systems with tunable acoustic properties. This evolving understanding has opened avenues for engineered phononic structures, seismic metamaterials, and advanced acoustic devices, rooted in fundamental studies of granular wave propagation.

2.1 Introduction

Sound propagation in granular media represents a critical research domain in both fundamental physics and engineering applications, including soil mechanics, geophysics, pharmaceuticals, and civil infrastructure. Granular materials, comprising discrete macroscopic particles, exhibit complex behaviors due to their disordered nature, inhomogeneous structure, and nonlinear inter-particle interactions.

Unlike crystalline solids, where wave propagation is relatively well understood, granular assemblies challenge classical models due to their heterogeneous and anisotropic nature. Understanding wave transmission through such media necessitates investigating the interactions between individual particles and their collective macroscopic responses.

This literature review presents a detailed chronological and thematic exploration of the research progress in this domain, covering theoretical advancements, experimental findings, and simulation-based models.

2.2 Timeline of Major Developments

- **1957 – Effective Medium Theory Initiated:** The earliest understanding of wave propagation in granular media relied on classical continuum mechanics and effective medium theories (EMTs). The Hertz-Mindlin contact theory, which describes the elastic behavior of two contacting spheres, became a foundational model in this field. Early EMTs assumed homogeneous, isotropic media and linear elasticity, ignoring key microstructural features of real granular systems such as force chains and contact anisotropy.

  Although useful, these models failed to explain experimental anomalies such as strong attenuation, dispersion, and the highly directional nature of wave paths in granular packings.

- **1992 – Sensitivity to Force Chains:** Experiments demonstrated that forces within a granular packing do not distribute uniformly but rather follow specific paths known as ”force chains.” These chains form a mechanically active network that carries the majority of stress through the medium, disrupting the assumptions of isotropy in classical theories.

  Liu and Nagel’s pioneering work revealed that acoustic waves travel predominantly along these force chains, and their arrangement critically affects sound propagation. These insights laid the foundation for considering granular media as inherently heterogeneous and dynamic systems.

- **1999 – Dual Nature of Propagation:**Experimental advancements enabled detailed acoustic measurements in three-dimensional granular packings. A seminal discovery was the coexistence of:

  - Ballistic, coherent waves: Representing an effective continuum response.
  - Coda-like, multiply scattered waves: Caused by internal heterogeneities and random contact orientations.

These observations were critical in identifying the dual nature of sound propagation in granular media, whereby low-frequency coherent pulses propagate through the medium's skeleton, while high-frequency signals undergo multiple scattering.

- **2006 – Microstructural Effects via DEM:** Using discrete element method (DEM) simulations, researchers investigated the dynamics of wave propagation in idealized granular systems, including regular and perturbed packings. Key findings included:
  - Anomalous acceleration of acoustic waves near the source due to contact force preloading and local anisotropy.
  - Sensitivity of wave velocity and signal shape to particle friction and polydispersity.

  These studies emphasized that even slight deviations from perfect ordering can significantly affect sound transmission, challenging any reliance on purely theoretical continuum approximations.

- **2007 – Experimental Validation of EMTs:** Advanced acoustic experiments conducted on dense packings of glass beads under varying stress conditions confirmed that:
  - The leading coherent signal corresponds well with EMT predictions under isotropic loading.
  - Subsequent signal components are dominated by scattered, high-frequency content reflecting local inhomogeneities.

  This provided strong experimental support for using a hybrid model of wave propagation: EMTs for coherent waves and multiple scattering theories for incoherent components.

- **2008–2009 – Elasto-Plastic and Cohesive DEM Modeling:** Mouraille and colleagues developed high-fidelity DEM simulations incorporating adhesive, frictional, and elasto-plastic contact models. Their work revealed:
  - The structure of granular packings is highly sensitive to contact laws used during sample preparation.
  - Despite changes in external loading and strain (elastic, failure, critical flow), the frequency spectrum of transmitted signals remained robust.
  - Simulation data offered critical insights into microstructural behavior, enabling interpretation of macroscale wave phenomena.

  This was a pivotal moment for computational modeling, showing that micro-contact mechanics significantly influence bulk acoustic properties.

- **2010 – Theoretical Modeling of Scattered Waves:** As computational power increased, the focus shifted toward bridging the gap between particle-level interactions and continuum-scale behavior. Researchers explored:
  - How micro-contact parameters (stiffness, adhesion, coordination number) influence emergent properties like elastic moduli and wave velocity.
  - Development of continuum models informed by microscale simulations and experiments.

These efforts aimed to formulate predictive frameworks capable of describing granular behavior without needing to simulate every individual grain.

- **2021–2022 – Hybrid DEM and Continuum Models:** Recent work has introduced hybrid computational frameworks that couple DEM with continuum solvers, such as finite-difference and finite-element methods. Notable advancements include:
  - Hybrid DEM–Finite Difference Models: Capable of simulating wave propagation through one-dimensional soil columns under seismic loading, including pore pressure buildup and liquefaction.
  - Poromechanics Integration: Applying Darcy's law to model fluid-solid interactions in saturated soils, providing more accurate representations of cyclic mobility and effective stress changes during seismic events.
  - Validation with Centrifuge Experiments: Models were calibrated and validated against high-fidelity physical experiments, reinforcing their reliability for predicting real-world behavior.

  These methodologies represent a significant step forward, enabling realistic simulations of granular media subjected to dynamic, site-specific, and multiphysical conditions.

2.3 Current Understanding

Over the years, The field has matured from simplified continuum theories to sophisticated multi-physics simulations and several theoretical and computational models have been developed to better understand sound transmission in granular media. These models aim to capture different aspects of wave behavior, from macroscopic approximations to microscale interactions. The most influential among them include:

- **Multiple Scattering Models:**

  Granular materials exhibit complex acoustic behaviors due to their inherent structural disorder, discrete contact networks, and non-uniform force transmission. Classical continuum models, while effective in certain regimes, often fail to capture the intricacies of wave propagation in granular media—especially in the presence of heterogeneities, scattering, and configuration-dependent effects. Multiple Scattering Theory (MST) has emerged as a compelling alternative framework, particularly suited for describing wave dynamics in such disordered systems.

  Theoretical work by Jia et al. (1999, 2004, 2009) and Trujillo et al. (2010) laid a comprehensive foundation for MST in the context of elastic waves in granular packings. This framework models the granular medium as an ensemble of randomly distributed scatterers and treats wave propagation as a statistical phenomenon dominated by multiple scattering events. By integrating concepts from statistical physics, quantum field theory, and elasticity, MST provides a rigorous approach to describe how coherent wavefronts decay into diffusive, multiply scattered waves—an essential aspect of sound transmission in granular systems.

  - **Importance of MST for Sound Transmission in Granular Media**

Modeling Structural Disorder and Scattering. MST directly addresses the challenges of disorder by treating the medium as a random field characterized by fluctuations in density and elastic moduli. The theoretical formalism accounts for scattering at grain contacts, local stiffness variations, and random force chains, which are often neglected in continuum theories.

Capturing Ballistic and Diffuse Regimes. A key strength of MST lies in its ability to model both the ballistic regime (initial coherent propagation) and the diffuse regime (long-time, incoherent wave scattering). This transition is central to understanding energy dissipation and wave attenuation in granular materials, particularly under stress or external perturbations.

Green's Function and Ensemble Averaging. MST utilizes ensemble-averaged Green's functions to represent the statistical wave response in disordered systems. The Dyson equation captures the average wave field, while the Bethe–Salpeter equation models intensity and energy transport, allowing for a detailed characterization of wave attenuation and spatial diffusion.

Bridge to Diffusion Theory. In the long-time limit, MST converges toward a diffusion equation for the wave energy, consistent with experimental observations of coda wave behavior. This provides a clear theoretical link between microscopic scattering processes and macroscopic transport laws like the Radiative Transport Equation (RTE).

Quantitative Predictability. The MST framework offers predictive expressions for quantities such as:

* Scattering mean free paths
* Extinction times
* Diffusion coefficients
* Mode conversion rates

These parameters are essential for analyzing wave transport, designing experiments, and interpreting signal attenuation in granular materials.

Multiple Scattering Theory (MST) offers a powerful and elegant framework for understanding sound transmission in granular materials. By embracing the statistical nature of disordered systems, MST moves beyond continuum approximations and captures essential features such as multiple scattering, coherence loss, and energy diffusion. The model's theoretical robustness and compatibility with experimental findings make it a valuable tool in fields ranging from geophysics and non-destructive testing to the study of acoustic metamaterials.

Nevertheless, MST has limitations—particularly in its assumptions of linearity and weak disorder, as well as its abstraction from specific microstructural dynamics. Future research integrating MST with nonlinear granular elasticity, higher-order scattering corrections, and high-resolution experimental validation will further enhance its predictive power and applicability to real-world granular systems.

- **Effective Medium Theory (EMT):** Sound propagation in granular materials offers a powerful, non-invasive method to probe their mechanical state and internal microstructure. As a disordered assembly of macroscopic particles, granular materials pose significant challenges for classical theories of continuum mechanics. Their behavior under stress—particularly the transmission of elastic waves—differs significantly from that of conventional solids due to factors such as nonlinearity, grain-scale heterogeneity, evolving contact networks, and the absence of thermal fluctuations. Understanding acoustic wave behavior in such systems is not only of theoretical interest but also has practical relevance in geophysics, civil engineering, and materials science.

  In granular assemblies, the speed of sound is not a fixed property but varies with the external stress, grain stiffness, and the evolving contact network. Classical Effective Medium Theory (EMT), which builds upon Hertz-Mindlin contact laws and assumes affine deformation fields and fixed coordination number, predicts that the bulk and shear moduli scale with pressure as K,μ p1/3, and hence the wave speeds scale as vs,vp p1/6. However, experimental observations often reveal stronger pressure dependence, particularly for the shear modulus and shear wave speed, indicating fundamental limitations in EMT. Makse et al. conducted ultrasonic wave experiments on granular packs under isotropic compression and found that both compressional and shear wave speeds increase faster than EMT predictions. These findings were further examined in their 2004 study, which combined experiments with numerical simulations and analytical theory to show that the primary source of discrepancy lies in the non-affine relaxation of grain motions under shear. While compressional deformations preserve the affine approximation reasonably well, shear deformations induce local rearrangements and rotations of grains, reducing the effective stiffness below EMT estimates.

  These findings underscore the need to move beyond classical EMT and explore frameworks that incorporate microstructural heterogeneity, evolving force chains, and viscoelastic relaxation. The study of sound propagation, thus, serves as a diagnostic for the internal state of granular media and reveals critical information about their rigidity, stability, and phase behavior near the jamming transition

  The propagation of sound in granular materials serves as a sensitive probe of their complex internal dynamics, revealing details about contact networks, mechanical stability, and proximity to critical transitions such as jamming. This review, based on the studies by Makse et al., has shown that while classical Effective Medium Theory (EMT) provides useful baseline predictions, it fails to account for key features observed in both experiments and simulations—particularly the nonlinear scaling of shear wave velocity and the role of non-affine relaxation. The inadequacy of EMT stems from its oversimplified assumptions: affine deformation, fixed coordination number, and isotropic contact distribution. As demonstrated, granular packings are inherently disordered systems with evolving microstructures, especially under low-pressure or shear-dominated conditions. Non-affine grain rearrangements, pressure-dependent coordination, and heterogeneous force transmission contribute significantly to the elastic response and acoustic behavior.

- **Force Chain Models:** Granular materials exhibit complex mechanical behavior due to their discrete structure, disordered geometries, and contact-driven interactions. Traditional continuum models often fall short in capturing the subtleties of such materials, particularly in the prediction of their elastic properties and wave propagation behavior. To bridge this gap, discrete models that incorporate inter-grain interactions—such as the Discrete Element Method (DEM) introduced by Cundall and Strack—have gained

prominence. These models simulate normal, shear, and sometimes bending interactions between grains. However, many standard approaches neglect angular (rotational) interactions, which are crucial for capturing rotational degrees of freedom and realistic mechanical responses in disordered media.

This paper extends these traditional models by introducing a Kirkwood-type spring network model that includes both central (normal) and angular interactions on a Delaunay tessellation of grains. The novelty lies in the inclusion of angular springs that can account for variations in Poisson's ratio—including the prediction of negative Poisson's ratios (auxetic behavior)—even when all individual grains have positive Poisson's ratios. This is particularly important in understanding and manipulating **sound transmission and attenuation in granular media, as auxetic materials often exhibit unique acoustic responses such as enhanced energy absorption and directional sound propagation.

The model is grounded in principles from solid-state physics but is generalized to disordered structures using Voronoi and Delaunay representations. In doing so, it provides a more realistic and computationally manageable framework for exploring how granular arrangements influence macroscopic mechanical and acoustic behavior. For sound transmission, the relevance of this model becomes apparent because wave propagation—especially shear and compressional waves—is intimately linked with the effective elastic moduli of the medium. By tuning the stiffness ratio between angular and central springs, one can simulate various degrees of grain-level constraints, influencing how vibrations travel through the medium.

- **Importance of Forced Chain Model for Sound Transmission in Granular Media**

  Captures Nonclassical Behavior: The model allows the simulation of auxetic behavior, which significantly affects how sound waves travel—potentially leading to bandgap formation or anomalous dispersion in granular phononic materials.

  Tunable Parameters: By varying the angular stiffness, the model offers a knob to modulate mechanical properties and therefore tailor acoustic behavior, making it a powerful tool in designing granular-based sound absorbers or insulators. .

  Elastic Wave Propagation: In the 2006 paper, the authors examine longitudinal and transverse sound velocities as a function of confining pressure. They find good agreement with Hertz-Mindlin contact theories, thereby validating the model's applicability in predicting acoustic properties under different loadings.

The Kirkwood-type spring network model proposed in this paper marks a significant advance in modeling the elastic behavior of disordered granular materials. Its ability to simulate auxetic effects and finely tune effective moduli offers valuable insights into wave propagation phenomena in granular media. Despite simplifications and limitations in shear modulus predictions, the model lays a strong foundation for future work, especially in the realm of acoustic metamaterials, granular dampers, and advanced composites.

- **Discrete Element Method (DEM):** Granular materials exhibit a rich array of mechanical responses due to the discrete nature of their constituent particles and the complex network

of contact forces. While continuum models capture macroscopic behavior effectively in some cases, they often fail to represent localized effects such as stress heterogeneity, force chain networks, and anisotropy in stress distribution. To address these limitations, Discrete Element Method (DEM) models have gained significant attention. These models allow simulation of individual particle interactions—including translational and rotational motion—by incorporating contact laws that govern force and torque transmission.

The works of Mouraille et al. (2006, 2009) provide critical advancements in DEM for modeling granular media by bridging micro-level particle interactions with macro-scale acoustic and mechanical properties. The 2006 paper investigates the sound propagation in granular materials using DEM by modeling grains as soft, frictional spheres with viscoelastic normal and tangential contacts. The 2009 study extends this by exploring wave transmission and effective moduli in both disordered and crystalline granular packings, highlighting the role of contact stiffness and microstructural arrangement on acoustic wave propagation.

Together, these papers advance our understanding of how local contact mechanics (e.g., normal/tangential stiffness, damping) and particle configurations affect global wave dynamics—key insights for applications in non-destructive testing, geotechnical engineering, and phononic metamaterials.

- **Importance of DEM for Sound Transmission in Granular Media**

  Particle-Scale Realism: The DEM framework used in both studies captures realistic inter-particle forces and allows direct modeling of wave transmission mechanisms such as scattering, attenuation, and localization due to particle-level disorder.

  Linking Micro to Macro: The models demonstrate how macroscopic elastic moduli—bulk and shear moduli—emerge from local interactions and evolve with coordination number, packing fraction, and stiffness ratios. This is crucial for understanding and engineering materials for specific acoustic properties.

  Elastic Wave Propagation: In the 2006 paper, the authors examine longitudinal and transverse sound velocities as a function of confining pressure. They find good agreement with Hertz-Mindlin contact theories, thereby validating the model's applicability in predicting acoustic properties under different loadings.

  Disorder and Crystallinity Effects: The 2009 study contrasts disordered packings with face-centered cubic (FCC) crystalline structures. It reveals that disorder leads to more isotropic but attenuated wave transmission, whereas crystalline systems support more defined modes with lower attenuation—an insight valuable in designing acoustic materials.

The DEM-based models proposed by Mouraille et al. mark a substantial contribution to our understanding of sound transmission in granular materials. By explicitly modeling particle-scale interactions and linking them to bulk acoustic properties, these studies provide a bridge between micro-mechanical modeling and macro-scale wave behavior. Despite computational limitations and simplifications in damping/contact models, the

framework remains a powerful tool for predicting and engineering wave propagation in granular systems. Future work incorporating better damping models, larger system sizes, and experimental comparisons will help validate and extend the findings, particularly for applications in acoustic metamaterials, geophysical sensing, and material design.

Each of these models contributes unique insights into the mechanisms governing sound transmission in granular materials. In the following sections, we provide a detailed review of their development, assumptions, strengths, and limitations.

3 Methods

3.1 Multiple Scattering Theory (MST)

Multiple Scattering Theory (MST) provides a rigorous framework for understanding the transition of sound waves from ballistic propagation to a multiply-scattered, diffuse regime in disordered granular media. This section outlines both the mathematical formulation and experimental observations supporting MST in the context of granular sound transmission [15, 27, 29, 10, 25].

3.1.1 Mathematical Formulation of MST

- **Elastic Wave Equation in Random Media:**
  - The motion of sound waves in granular media is governed by the following equation:

$$\rho(\vec{r})\frac{\partial^2 u_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + f_i(\vec{r}, t)$$

    The term on the right-hand side represents the divergence of the stress tensor, which is responsible for the wave propagation, and the external forces $f_i(\vec{r}, t)$, which can include any external perturbations acting on the granular medium [27].
  - The stress tensor σ_{ij} that characterizes the mechanical response of the medium is defined as:

$$\sigma_{ij} = \lambda(\vec{r})\delta_{ij}\nabla \cdot \vec{u} + 2\mu(\vec{r})\epsilon_{ij}$$

    The first term, involving the Lamé parameter $\lambda(\vec{r})$, corresponds to the volumetric (hydrostatic) response of the material, while the second term, involving the Lamé parameter $\mu(\vec{r})$, represents the deviatoric (shear) response. Here, $\nabla \cdot \vec{u}$ is the volumetric strain and ϵ_{ij} is the strain tensor, which characterizes the local deformation of the medium.

- **Jiang–Liu Nonlinear Elasticity Model:**
  - The Jiang–Liu model captures the nonlinear elasticity of granular media by introducing strain-dependent moduli. These moduli are represented as:

$$K = \tilde{K}\delta^b, \quad G = \tilde{G}\delta^a, \quad \delta = -\epsilon_{ll}$$

The exponents a and b control the nonlinear dependence of the moduli on the compression δ, reflecting the compressive behavior of granular media [27].

- The free energy density $F(\epsilon)$ is a function of the strain components and is given by:

$$F(\epsilon) = \frac{1}{2}K\epsilon_{ll}^2 + G\epsilon_s^2$$

Here, ϵ_{ll} is the longitudinal strain, and ϵ_s^2 is the deviatoric strain, which is defined as:

$$\epsilon_s^2 = \epsilon_{ij}^0\epsilon_{ij}^0, \quad \epsilon_{ij}^0 = \epsilon_{ij} - \frac{\epsilon_{ll}}{3}\delta_{ij}$$

This formulation accounts for both the volumetric and deviatoric components of the strain, capturing the complexity of granular media's mechanical response under deformation.

- **Green's Function and Dyson Equation:**

  - In order to analyze wave propagation in disordered granular media, the Green's function $G(\omega)$ is used to describe the response of the medium to a point source. The resolvent operator for the Green's function is defined as:

$$G(\omega) = [\omega + i\epsilon - \mathcal{K}]^{-1}$$

where ω is the frequency of the wave, $i\epsilon$ represents a small imaginary part that ensures proper convergence, and $\mathcal{K}$ is the wave operator that accounts for the medium's properties and scattering effects [27].

  - The Dyson equation provides an ensemble-averaged Green's function, which incorporates the scattering effects through the self-energy $\Sigma(\omega)$:

$$\langle G(\omega)\rangle = \left[G_0^{-1}(\omega) - \Sigma(\omega)\right]^{-1}$$

where $G_0(\omega)$ is the Green's function for a homogeneous medium, and $\Sigma(\omega)$ represents the self-energy, which is a measure of the scattering interactions in the disordered granular medium. The self-energy is related to the Green's function through:

$$\Sigma \approx \langle \delta\mathcal{K} \cdot G_0 \cdot \delta\mathcal{K}\rangle$$

This equation provides a key tool for understanding how multiple scattering events affect wave propagation in a random medium.

- **Bethe–Salpeter Equation and Radiative Transport:**

  - The Bethe–Salpeter equation describes the second-order correlation of the Green's function, which provides the wave intensity L:

$$L = \langle G \otimes G^*\rangle$$

This second-order moment of the Green's function describes the intensity of waves propagating through the medium and accounts for multiple scattering effects [25, 10].

- The Bethe–Salpeter equation is written as:

$$L = G \otimes G^* + G \otimes G^* : U : L$$

where U is the irreducible vertex function that captures the irreducible scattering interactions between waves. This equation describes how wave intensity is influenced by scattering and interactions within the granular medium.

- In the limit of quasi-isotropic scattering, the Radiative Transport Equation (RTE) can be derived:

$$\left(\frac{\partial}{\partial t} + \vec{v}j \cdot \nabla + \frac{1}{\tau_j}\right) L_j(\vec{x}, t) = S_j \delta(\vec{x})\delta(t) + \sum j' \int W_{jj'} L_{j'} d\Omega'$$

This equation describes the propagation and scattering of wave intensity in the medium, where τ_j is the scattering time, and $W_{jj'}$ is the scattering kernel that governs the scattering between different directions. At long timescales, the RTE simplifies to the diffusion equation:

$$\partial_t I = D\nabla^2 I - \frac{I}{\tau_a}$$

where D is the diffusion coefficient, ∇^2 is the spatial Laplacian, and τ_a is the attenuation time. This equation describes the diffusion of wave intensity due to multiple scattering in the medium.

3.1.2 Experimental Validation of MST in Granular Media

- **Sample Configuration and Loading:**
  - The granular medium used in the experiments consisted of polydisperse glass beads with diameters ranging from 0.6 to 0.8 mm, packed inside a duralumin cylinder with a diameter of 30 mm. The granular material was preloaded using pistons to apply uniaxial compressive stress up to 0.75 MPa.
  - A loading-unloading cycle was performed to stabilize the contact network within the granular medium and minimize hysteresis, ensuring consistent wave propagation during the measurements[15].
- **Ultrasound Excitation and Detection:**
  - A 30 mm transducer was used to apply a 500 kHz longitudinal ultrasonic burst (10 cycles, 20 µs duration), which generated sound waves with wavelengths comparable to the particle size ($kd \approx 2.9$). This excitation allowed both coherent and scattered waves to be generated in the granular medium[15].
  - A 2 mm receiver at the bottom of the cylinder recorded the transmitted signals, allowing for the study of both the ballistic and scattered components of wave propagation as a function of stress and particle arrangement within the granular medium.
- **Coherent Pulse and Coda Analysis:**

- The transmitted signal was analyzed to separate the coherent pulse from the trailing coda. The coherent pulse represents the ballistic wave propagation, which self-averages over different configurations, while the coda represents high-frequency, multiply scattered waves that vary depending on the packing geometry.
- Averages were taken over 15–50 independent packings to separate the coherent and non-coherent parts of the signal, providing insight into the scattering effects and wave attenuation in the medium [15, 10].

- **Linking Observations to Theory:**
  - The velocity of the coherent front observed in the experiments matched the predictions from the Jiang–Liu elasticity model, confirming the validity of the nonlinear elasticity description of granular media [15, 29].
  - The decay of the coda was fitted to the diffusion solution of the Radiative Transport Equation (RTE), allowing for the estimation of key parameters such as the diffusion coefficient $D = 0.13\,\mathrm{m}^2/\mathrm{s}$, the quality factor $Q^{-1} = 0.005$, and the scattering length $\ell \approx$ few mm.
  - These experimental results provided quantitative validation of the MST framework, demonstrating its applicability in modeling wave propagation in disordered granular media.

3.2 EMT

- **Overview:** Studies of sound propagation in granular materials rely on two complementary approaches: controlled laboratory experiments and numerical simulations using discrete element methods (DEM). These methods provide insight into wave dynamics, elastic moduli, and the evolution of contact networks under stress [21].

- **Experimental Setup (Makse et al.):**
  - Granular assembly of glass beads under isotropic compression inside a pressure vessel.
  - Ultrasonic pulses transmitted and received via acoustic transducers at opposite ends.
  - Time-of-flight measurements used to determine compressional and shear wave velocities:
$$v_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}, \quad v_s = \sqrt{\frac{\mu}{\rho}}$$
  - Experiments across varying pressures revealed that shear modulus $\mu(p) \sim p^{2/3}$, deviating from the $p^{1/3}$ predicted by effective medium theory (EMT) [21].

- **Numerical Simulations (Molecular Dynamics):**
  - Particles obey Newton's laws of motion:
$$m_i \frac{d^2\vec{r}_i}{dt^2} = \sum_j \left(\vec{F}^n_{ij} + \vec{F}^t_{ij} + \vec{F}^d_{ij}\right)$$

- Normal force via Hertzian contact:

$$F_n = \frac{2}{3} k_n R^{1/2} \delta^{3/2}$$

- Tangential force via Mindlin model:

$$F_t = k_t R^{1/2} \delta^{1/2} s$$

- Simulations under varying pressure studied changes in coordination number Z, stress networks, and moduli. Shear modulus was especially sensitive to non-affine relaxation dynamics [21].

- **Hertz-Mindlin Contact Model:**
  - Describes contact forces between elastic spheres under small deformations.
  - Contact radius and normal force:

$$a = \left(\frac{3F_n R}{4E^*} \right)^{1/3}, \quad F_n = \frac{4}{3} E^* R^{1/2} \delta^{3/2}$$

  - Effective Young's modulus:

$$\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

  - Tangential force (Mindlin model) [21, 16]:

$$dF_t = 8G^* a \, ds = k_t \delta^{1/2} ds$$

- **Biot's Theory of Wave Propagation in Porous Media:**
  - Models saturated granular media as a solid skeleton and a pore-filling fluid.
  - Coupled equations for solid $\vec{u}$ and fluid $\vec{U}$ displacements:

$$\rho_{11} \ddot{\vec{u}} + \rho_{12} \ddot{\vec{U}} = \nabla \cdot \sigma - \alpha \nabla p$$

$$\rho_{12} \ddot{\vec{u}} + \rho_{22} \ddot{\vec{U}} = -\nabla p - b(\vec{U} - \vec{u})$$

  - Predicts three wave modes: two compressional (fast and slow) and one shear.
  - Accounts for frequency-dependent attenuation and dispersion due to viscous coupling [5, 6].

- **Gassmann's Equation for Saturated Media:**
  - Assumes quasistatic limit with no relative fluid-solid motion.
  - Bulk modulus of saturated medium:

$$\frac{K_{\text{sat}} - K_{\text{dry}}}{K_s - K_{\text{dry}}} = \frac{K_f}{\phi K_s + (1-\phi) K_f}$$

  - Rearranged:

$$K_{\text{sat}} = K_{\text{dry}} + \frac{\left(1 - \frac{K_{\text{dry}}}{K_s}\right)^2}{\frac{\phi}{K_f} + \frac{1-\phi}{K_s} - \frac{K_{\text{dry}}}{K_s^2}}$$

  - Useful in interpreting acoustic response of fluid-saturated granular systems at low frequencies [11].

3.3 Force Chain

3.3.1 Spring Network with Central and Angular Interactions

- **Central Interactions:**
  - For a Delaunay graph $G(V, E)$, edge e connects two nodes with a central force [1, 24, 3, 14]:
$$F_i = k^n \hat{n}_{ij}(u_j - u_i)$$
  - The energy in a Voronoi cell becomes:
$$U_{central} = \frac{1}{2}\sum_{e=1}^{e_{\text{total}}} k^n (\Delta u^{(e)})^2$$
  - Assuming a regular hexagonal lattice, effective continuum moduli are:
$$C_{1111} = C_{2222} = \frac{9}{4}k^n, \quad C_{1122} = C_{2211} = \frac{3}{4}k^n, \quad C_{1212} = \frac{3}{4}k^n$$
  - These satisfy isotropy:
$$2C_{1212} = C_{1111} - C_{1122}$$
  - Bulk and shear moduli:
$$K = \frac{1}{2}(C_{1111} + C_{1122}) = k^n, \quad \mu = C_{1212} = \frac{3}{4}k^n$$
  - Planar Young's modulus and Poisson's ratio:
$$E = \frac{4K\mu}{K+\mu}, \quad \nu = \frac{K-\mu}{K+\mu}, \quad \text{with } \nu = \frac{1}{3} \text{ if only } k^n \text{ present}$$
  - **Figure:**
- **Angular Interactions:**
  - Angular springs of stiffness k^a are introduced between edge pairs.
  - Angular energy per Voronoi cell:
$$U_{angular} = \sum_{e=1}^{e_{\text{total}}} \frac{1}{2}k^a(\Delta\phi)^2$$
  - Total energy:
$$U = U_{central} + U_{angular}$$
  - Moduli with angular interactions:
$$K = k^n, \quad \mu = \frac{3}{4}(k^n + k^a)$$
  - Poisson's ratio becomes:
$$\nu = \frac{1 - 3(k^a/k^n)}{3 + (k^a/k^n)}, \quad \text{ranging from } \frac{1}{3} \text{ to } -1$$
- **Bending Interactions:**
  - In full DEM, bending is included:
$$U = U_{central} + U_{shear} + U_{bending}$$
  - Edges act like beams, and resulting continuum is micropolar [3].

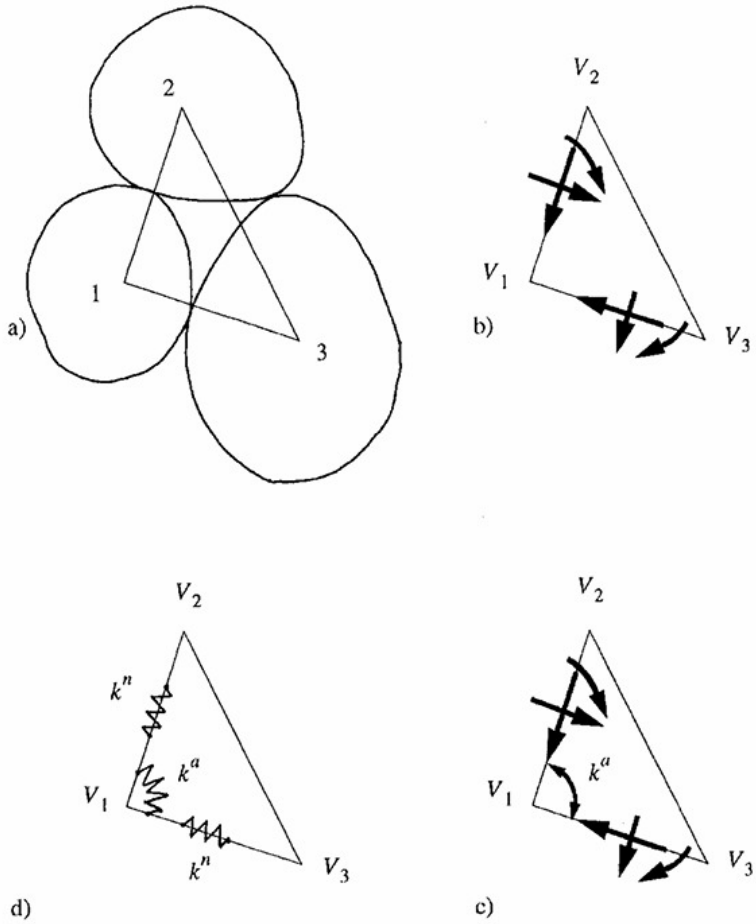

Fig. 1 (*a*) A cluster of three grains (1, 2, 3) showing the three lines of interactions; (*b*) a discrete element model showing the normal force, the shear force, and the moment exerted by grains 2 and 3 each onto the grain 1; (*c*) a most general model showing the same grain-grain interactions as before but augmented by an internal, angular spring constant k^a; (*d*) a simplified model adopted in this paper, showing only normal (k^n) and angular (k^a) effects

3.3.2 Two-Phase Granular Microstructure

- A disordered, planar granular medium is modeled by a Delaunay tessellation [24].
- Generated via Poisson point process in $\mathbb{R}^2$:

$$P\{N(A) = k\} = e^{-\lambda|A|}\frac{(\lambda|A|)^k}{k!}, \quad A \subset \mathbb{R}^2$$

- Angular stiffness ratio:

$$s_r = \frac{k^n l^2}{k^a}$$

- Grains are of two types: soft (k_1) and stiff (k_2), with contrast $r = \frac{k_2^n}{k_1^n}$ and volume fractions c_1, c_2.
- Unit cell subjected to:
  - Biaxial strain: $\varepsilon_{11} = \varepsilon_{22}$
  - Shear strain: $\varepsilon_{12} = -\varepsilon_{21}$
- Energy equivalence yields effective moduli.

3.3.3 Numerical Results and Effective Medium Models

- $s_r = \infty$: no angular stiffness; $s_r = 0.1$: dominant angular stiffness.
- Simulation for $s_r = \infty, 100, 50, 10, 1.0, 0.1$ tracks normalized moduli K/K_1, μ/μ_1 vs. c_2.
- **Figure:**
- Effective Poisson's ratio ν is convex in c_2 and can be negative even if phases have positive ν [8, 12, 26].
- Hill and Budiansky moduli formulas:

$$\frac{1}{\kappa} = \frac{1}{\kappa_1} + c_2\left(\frac{1}{\kappa_2} - \frac{1}{\kappa_1}\right)\frac{\frac{1}{\mu} + \frac{1}{\kappa}}{\frac{1}{\mu} + \frac{1}{\kappa_2}}$$

$$\frac{1}{\mu} = \frac{1}{\mu_1} + 2c_2\left(\frac{1}{\mu_2} - \frac{1}{\mu_1}\right)\frac{\frac{1}{\mu} + \frac{1}{\kappa}}{\frac{1}{\mu} + \frac{1}{\mu_2} + \frac{2}{\kappa}}$$

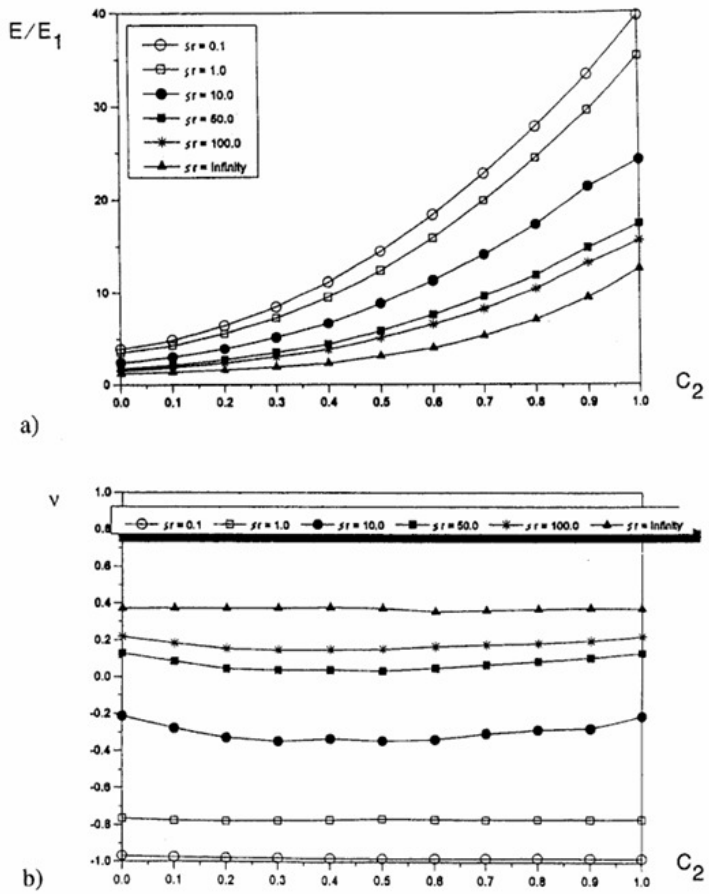

Fig. 4 Effective Young's modulus (*a*) and Poisson's ratio (*b*) at contrast 10 for the same range of values of *sr* as in Fig. 3

3.3.4 Effective Medium Approximations

- **Compared Models:**
  - Perfect Bonding (PB)
  - Symmetric Self-Consistent Approximation (SCA-S): treats all phases equally
  - Asymmetric Self-Consistent Approximation (SCA-A): treats one phase as matrix
- SCA-S conditions:

$$\sum_{i=1}^{N} c_i P_i (K - K_i) = 0, \quad \sum_{i=1}^{N} c_i Q_i (\mu - \mu_i) = 0$$

- SCA-A equations:

$$\frac{K - K_1}{K + A_i} = \sum_{i=2}^{N} c_i \left(\frac{K_i - K_1}{K_i + A_i} \right), \quad \frac{\mu - \mu_1}{\mu + B_i} = \sum_{i=2}^{N} c_i \left(\frac{\mu_i - \mu_1}{\mu_i + B_i} \right)$$

- Coefficients:

$$P_i = \left[1 + Z_i - \frac{Z_i Y_i}{X_i + Y_i} \left(\frac{1 - s}{1 + s} \right)^2 \right]^{-1}$$

$$2Q_i = P_i \frac{1 + Z_i}{X_i + Y_i} + (2 + A_i - X_i)^{-1}$$

- Parameters:

$$X_i = 1 + \frac{A_i s (1 + \nu)}{(1 + s)^2}, \quad Y_i = \frac{A_i (1 - \nu)}{2}, \quad Z_i = \frac{C_i (1 + \nu)}{2}$$

$$A_i = \frac{\mu_i}{\mu} - 1, \quad C_i = \frac{\kappa_i}{\kappa}, \quad \nu = \frac{\kappa - \mu}{\kappa + \mu}$$

- All models match K well, but show deviation in μ.

3.4 Discrete Element Method (DEM)

The Discrete Element Method (DEM) has become a central tool for simulating and understanding the dynamics of granular media [9, 20]. In DEM, the motion of individual particles and their interactions via contact forces are modeled explicitly. This section outlines the experimental setup in DEM, the contact force models, numerical integration, and data analysis strategies used to study sound propagation and stress wave transmission in granular materials [22].

3.4.1 DEM Setup in Granular Systems

- **Particle Assembly and Geometry:**
  - **Particle Characteristics:** Granular materials are modeled as spherical particles with a defined diameter (e.g., 0.5–1.0 mm), density (e.g., 2000 kg/m^3), and mass calculated via $m = \frac{4}{3}\pi \left(\frac{d}{2}\right)^3 \rho$ [22].

- **Packing Structure:** Particles can be arranged in either regular (e.g., face-centered cubic, FCC) or randomized configurations to simulate different levels of order. Polydispersity is introduced by assigning a slight size distribution to capture disorder effects [23].
- **Boundary Conditions:** Simulations typically employ periodic boundaries in the transverse (x and y) directions and fixed or free boundaries in the propagation (z) direction, depending on the desired experimental mimicry (e.g., confined vs. unconfined settings) [20, 28].

- **Loading and Excitation:**
  - **Quasi-static and Dynamic Loading:** Systems can be subjected to static or dynamic loads. For sound propagation studies, a small amplitude harmonic excitation (ultrasonic burst or continuous sine wave) is applied at one boundary.
  - **Stress and Strain Control:** In compression tests, a predefined strain or stress is applied through movable walls, allowing the investigation of mechanical responses under different loading regimes [22].

3.4.2 Advanced Contact Force Model in DEM

To accurately simulate mechanical wave propagation in dense granular packings, we adopt a physically motivated, history-dependent contact model developed by Mouraille and Luding [20]. The particles are modeled as non-deformable perfect spheres that interact only upon contact [9].

- **Contact Detection and Overlap:** Two particles i and j, with radii a_i and a_j, are considered in contact if the center-to-center distance $|\mathbf{r}_i - \mathbf{r}_j|$ is less than $a_i + a_j$. The normal overlap δ is defined as:
$$\delta = (a_i + a_j) - |\mathbf{r}_i - \mathbf{r}_j| > 0,$$
with the unit normal vector pointing from j to i:
$$\mathbf{n} = \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|}.$$

- **Normal Contact Force:** The total force on particle i is a sum of all pairwise forces from particles in contact:
$$\mathbf{f}_i = \sum_j \mathbf{f}_{ij}^c,$$
where $\mathbf{f}_{ij}^c = f_n \mathbf{n} + \mathbf{f}_t$ is decomposed into normal and tangential components.

  The normal force f_n is modeled using an adhesive, elasto-plastic, history-dependent law. It depends on: - the current overlap δ, - the relative surface velocity in the normal direction $\dot{\delta}$, - the maximum historical overlap δ_{max} during contact.

  During the initial loading, the normal force follows:
$$f_n = k_1 \delta + \gamma_n \dot{\delta},$$
where k_1 is the initial loading stiffness, and γ_n is the viscous damping coefficient [22, 7].

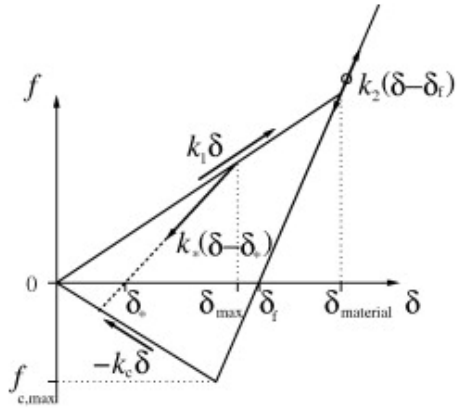

Figure 1: Illustration of the spring component of the normal contact force f_n, which depends solely on the overlap δ. Viscous damping forces in the normal direction are not included in this plot. The constants k_1, k_2, k_c, and δ_{material} are material-specific parameters. The quantity δ_{max} represents the maximum overlap experienced during the contact's history and introduces history dependence into the force law. The overlap δ_* denotes the point at which the force becomes zero under loading or unloading using the current effective stiffness k_*. At large overlaps, the force becomes zero at δ_f when the maximum stiffness k_2 is applied.

Upon unloading and reloading, δ_{max} is retained as a history variable. The force evolves piecewise-linearly: - A stiffness $k(\delta)$ is interpolated between k_1 and a higher stiffness k_2, based on δ_{max}. - In the adhesive regime (negative force), the maximum attraction is limited by $k_c\delta$, where k_c is the cohesion stiffness.

Thus, the full force law is:

$$f_n = \begin{cases} k_1\delta + \gamma_n\dot{\delta} & \text{(initial loading)} \\ k(\delta)\delta + \gamma_n\dot{\delta} & \text{(unloading/reloading)} \\ -k_c\delta & \text{(maximum adhesion limit).} \end{cases}$$

- **Tangential Contact Force:** Tangential forces arise from tangential sliding and relative rotation of the contacting particles. The tangential velocity [28] at contact is given by:

$$\mathbf{v}_t = \mathbf{v}_{ij} - (\mathbf{v}_{ij} \cdot \mathbf{n})\mathbf{n},$$

where the relative surface velocity is:

$$\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j + a_i^0(\mathbf{n} \times \boldsymbol{\omega}_i) + a_j^0(\mathbf{n} \times \boldsymbol{\omega}_j),$$

and $a_i^0 = a_i - \delta/2$, $a_j^0 = a_j - \delta/2$.

The tangential force is computed as:

$$\mathbf{f}_t = -k_t\boldsymbol{\xi} - \gamma_t\mathbf{v}_t,$$

where k_t is the tangential stiffness, $\boldsymbol{\xi}$ is the tangential displacement (accumulated over contact duration), and γ_t is the tangential damping.

Coulomb's law restricts the maximum tangential force:

$$|\mathbf{f}_t| \leq \mu|f_n|,$$

with μ the friction coefficient.

- **Background Damping (Viscous Environment):** To dissipate energy associated with long-wavelength cooperative motion that is not effectively damped by two-body contacts, additional background damping is introduced:

$$\mathbf{f}_i = \sum_j (\mathbf{f}_n + \mathbf{f}_t) - c_b \mathbf{v}_i, \quad \boldsymbol{\tau}_i = \sum_j \boldsymbol{\tau}_{\text{friction}} - c_{br} a_i^2 \boldsymbol{\omega}_i,$$

  where c_b and c_{br} are the translational and rotational background viscosity coefficients, respectively. These act as effective damping from an interstitial viscous medium and accelerate system relaxation. Care must be taken to avoid over-damping [23].

- **Units and Parameters:**

Property	Symbol	Value	Dimensional units	SI units
Time unit	t_u	1	100 µs	10^{-4} s
Length unit	x_u	1	1 mm	10^{-3} m
Mass unit	m_u	1	1 mg	10^{-6} kg
Particle radius	a_0	0.005	5 µm	5×10^{-6} m
Material density	ρ	2	2 mg/mm^3	2000 kg/m^3
Max. loading/unloading stiffness	k_2	5	5 mg/(100 µs)^2	5×10^2 kg/s^2
Initial loading stiffness	k_1/k_2	0.5		
Adhesion "stiffness"	k_c/k_2	0.2		
Friction stiffness	k_t/k_2	0.2		
Coulomb friction coefficient	$\mu = \mu_d = \mu_s$	1		
Dynamic to static friction ratio	$\phi_d = \mu_d/\mu_s$	1		
Normal viscosity	$\gamma = \gamma_n$	5×10^{-5}	5×10^{-5} mg/100 µs	5×10^{-7} kg/s
Tangential viscosity	γ_t/γ	0.2		
Background viscosity	γ_b/γ	4.0		
Background viscous torque	γ_{br}/γ	1.0		
Fluid overlap	ϕ_f	0.05		

Figure 2: Microscopic material parameters used (third column), if not explicitly specified. The fourth column contains these values in the dimensional units, i.e., when the time-, length-, and mass-units are $100\ \mu$s, mm, and mg, respectively.

3.4.3 Numerical Integration and Simulation Details

- **Time Stepping:**
  DEM simulations typically use explicit time integration schemes (e.g., Velocity-Verlet). The time step Δt must be much smaller than the collision duration:

$$\Delta t < C\sqrt{\frac{m}{k_n}},$$

  where C is a safety factor (commonly 0.1 to 0.2).

- **Solver Techniques:**
  For steady-state and frequency-domain analysis, the equations of motion can be linearized and solved using iterative methods such as the Bi-Conjugate Gradient Method, especially when dealing with large, sparse systems.

3.4.4 Data Analysis: Wave Propagation and Dispersion

- **Fourier Analysis:**
  The time-dependent displacement or stress data are transformed using the Fast Fourier Transform (FFT) to extract frequency content and dispersion relations $\omega(k)$ [23].

- **Phase and Group Velocity Calculation:**
  The phase shift of the wave at various positions allows extraction of phase velocity v_ϕ and group velocity v_g, typically given by:

$$v_\phi = \frac{\omega}{k}, \quad v_g = \frac{d\omega}{dk}.$$

- **Scattering and Diffusion:**
  In heterogeneous media, spatial disorder causes scattering of acoustic energy. Analysis of the coda waves and power spectra (e.g., following a f^{-psi} law) provides insights into energy diffusion characteristics [22].

3.4.5 Connections to Macroscopic Models

DEM results are often used to validate or calibrate macroscopic continuum models [22]:

- **Nonlinear Elasticity Models:**
  Experimental observations on confined granular packings are often compared with predictions from models such as the Jiang–Liu model for nonlinear elasticity:

$$\epsilon_{ij} = u_{ij} + u_{ij}^p \approx u_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right),$$

$$F(\epsilon) = \frac{1}{2}K\epsilon_{ll}^2 + G\epsilon_s^2,$$

  where K and G are pressure-dependent bulk and shear moduli, respectively.

- **Wave Equation in Random Media:**The macroscopic wave motion is governed by:

$$\rho(r)\frac{\partial^2 u_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + f_i(r,t),$$

  with the stress tensor given by:

$$\sigma_{ij}(r) = \lambda(r)\delta_{ij}\epsilon_{kk} + 2\mu(r)\epsilon_{ij}.$$

  This formulation accounts for scattering due to spatial fluctuations in density and elasticity, linking DEM-measured microstructural parameters to effective macroscopic behavior.

- **Radiative Transport and Diffusion:**For strongly scattering media, the Radiative Transport Equation (RTE) and its diffusion approximation become relevant. The energy density U_i satisfies:

$$\partial_t U_i - \nabla \cdot \left(\sum_j D_{ij}\nabla U_j\right) = S_i - \sum_j C_{ij}U_j - \frac{U_i}{\tau_{a,i}},$$

  where D_{ij} is the diffusion tensor, C_{ij} characterizes scattering losses, and $\tau_{a,i}$ is the absorption (or relaxation) time. The parameters D_{ij}, C_{ij}, and $\tau_{a,i}$ can be fitted using the decay of coda waves observed in the simulations.

3.5 Hybrid Numerical–Experimental Framework model

Saturated granular media present significant challenges for experimental resolution, particularly in the study of high-frequency wave propagation [18]. Conventional setups—such as shear cells, inclined planes, and Couette devices—primarily operate in low-frequency, quasi-static regimes. While useful for studying frictional rheology and strain localization, they are ill-suited for capturing transient stress redistribution, wave speed variation, and pore pressure dynamics [2].

To overcome these limitations, we adopt a hybrid multiscale numerical framework that couples a continuum-scale finite difference model with micro-scale Discrete Element Method (DEM) simulations [18, 17]. This approach enables resolution of time-dependent, scale-coupled dynamics such as stress wave transmission, poroelastic attenuation, and fluid–solid interactions across multiple temporal and spatial scales.

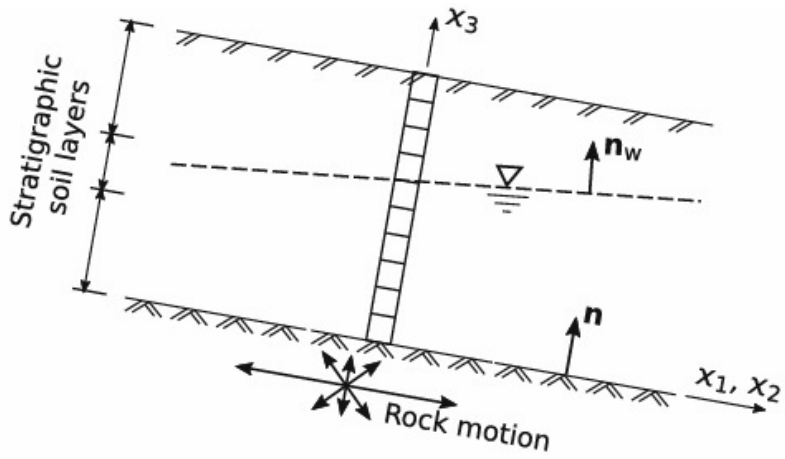

Fig. 1 One-dimensional problem of wave propagation through soil.

3.5.1 Limitations of Conventional Laboratory Techniques

The following experimental configurations illustrate the capabilities and constraints of typical granular flow setups in relation to high-frequency acoustic phenomena:

1. **Shear Cell Experiments**
   - **Setup:** Parallel plates or Couette geometry with a moving boundary to impose shear rate $\dot{\gamma}$. Normal stress P is recorded using load cells; shear stress τ is inferred from resistance.
   - **Purpose:** Evaluation of $\mu(I)$ rheology where $\mu = \tau/P$, and $I = \dot{\gamma} d\sqrt{\rho/P}$.
   - **Limitation:** Primarily suitable for steady-state rheology; lacks capability for high-speed dynamic measurements [18].
2. **Inclined Plane Experiments**
   - **Setup:** Granular flow down a plane at angle θ, controlled by grain size, slope, and layer height h.
   - **Measurements:** Surface velocity profiles, flow thickness, and flow thresholds.

- **Limitation:** Designed for steady flows; transient wave phenomena are heavily damped and unresolved [2].

3. **Annular Couette Cell**
   - **Setup:** Rotating inner cylinder with adjustable pressure via a lid or piston.
   - **Measurements:** Torque sensing, particle imaging.
   - **Limitation:** Operates under quasi-static conditions, limiting temporal resolution of wave dynamics.
4. **Split-Bottom Cell**
   - **Setup:** A rotating inner disk induces shear without wall effects.
   - **Measurements:** Force chains, granular trajectories, and local strain fields.
   - **Limitation:** Extremely low strain rates make it unsuitable for capturing fast wave phenomena [18].
5. **Refractive Index-Matched Imaging**
   - **Setup:** Fluid and particle refractive indices are matched for internal 3D tracking using laser or fluorescent imaging.
   - **Utility:** Enables visualization of rearrangements and shear layers.
   - **Limitation:** Limited temporal resolution; cannot resolve rapid stress wave propagation [19].

3.5.2 Need for a Mathematical and Numerical Framework

Despite their utility in understanding granular mechanics and frictional rheology, experimental setups fall short in capturing the transient behavior central to wave propagation. The governing dynamics involve time-dependent fields such as displacement, strain rate, and pore pressure—all evolving on millisecond to microsecond scales [4].

3.5.3 Experimental Limitations

- **Shear and Couette cells** allow study of $\mu(I)$ rheology but do not resolve dynamic redistribution of stresses governed by momentum conservation.
- **Inclined plane and split-bottom setups** capture strain localization and flow thresholds, but not pore pressure diffusion or elastic wave transmission [13].
- **DEM-specific phenomena** such as:
  - Inter-particle contact dynamics,
  - Viscous damping at micro-contacts,
  - Nonlinear Hertzian contact forces,

  are not accessible via physical measurements.

3.5.4 Numerical Techniques

3.5.5 Macro-Scale Continuum Model

The one-dimensional poromechanical behavior is described using Biot's formulation for saturated media [4]:

1. **Solid Phase Motion**

$$\rho \frac{\partial^2 u}{\partial t^2} = \frac{\partial \sigma}{\partial z} + \rho g$$

2. **Pore Pressure Evolution**

$$\frac{\partial p}{\partial t} = \frac{1}{n\beta_w + \alpha} \left[\frac{\partial}{\partial z} \left(k \frac{\partial p}{\partial z} \right) - M \frac{\partial \varepsilon_v}{\partial t} \right]$$

These equations govern vertical stress transmission and pore pressure evolution and form the basis of the finite difference scheme [18].

3.5.6 Micro-Scale Discrete Element Simulations

Each spatial node (Representative Volume Element, RVE) is equipped with a DEM simulation to compute the local granular response:

- **Elastic Contacts:** $F_n = k_n \delta^{3/2}$
- **Viscous Damping:** $F_{\text{visc}} = -\eta \frac{d\delta}{dt}$

DEM outputs—volumetric strain, effective stress, and contact force distribution—inform the macro-scale equations.

3.5.7 Constitutive Coupling

To maintain poromechanical consistency, the total stress is decomposed as:

$$\sigma(z,t) = \sigma'(z,t) + p(z,t) \tag{1}$$

This formulation captures the transition to liquefaction, as $\sigma' \to 0$ under increasing pore pressure [2, 18].

3.5.8 Leapfrog Time Integration Scheme

A leapfrog finite-difference method advances the displacement and pressure fields in time [18]:

$$u_i^{n+1} = 2u_i^n - u_i^{n-1} + \Delta t^2 \left(\frac{\sigma_{i+1}^n - \sigma_i^n}{\Delta z} + \rho g \right) \tag{2}$$

$$p_i^{n+1} = p_i^n + \Delta t \left(\text{Darcy diffusion} - M \frac{\Delta \varepsilon_v}{\Delta t} \right) \tag{3}$$

3.5.9 Boundary and Initial Conditions

- **Bottom boundary:** Prescribed displacement simulates seismic input.
- **Top boundary:** Modeled as either a free surface or saturated interface.
- **Pore pressure:** Drained or undrained, depending on boundary drainage assumptions [19].

4 Discussion

Understanding sound transmission in granular media (GM) is inherently complex due to the dual phase nature (solid grains and interstitial fluid), nonlinear contact mechanics, multiscale interactions, and dissipative mechanisms. Although substantial advances have been made, modeling remains challenging, especially in capturing energy dissipation, grain-fluid coupling, and non-linear dynamic behavior. Addressing these gaps requires integrating insights from experimental observations, advanced numerical techniques, and rigorous theoretical frameworks.

Over time, various modeling approaches have emerged, each targeting specific facets of GM dynamics:

4.1 Multiple Scattering Theory (MST)

Simplified Assumptions and Granular System Complexity: Traditional MST frameworks rely on assumptions of weak scattering, homogeneous scatterer properties, and isotropic media. These approximations fail to capture the inherent complexity of granular systems, which exhibit strong heterogeneity, polydispersity, and anisotropic force networks. The discrete, dissipative nature of granular media introduces non-affine deformations and contact rearrangements that violate MST's continuum-based premises.

Neglect of Mode Conversion Effects: Most MST formulations ignore critical phenomena like mode conversion between compressional (P-wave) and shear (S-wave) modes. In disordered granular assemblies, such conversions dominate energy redistribution and attenuation due to rotational particle dynamics and force chain reconfigurations. This limitation restricts MST's predictive power for real-world granular systems where multi-mode interactions are intrinsic.

Isotropic Scattering vs. Anisotropic Reality: MST's assumption of isotropic scattering behavior conflicts with the directional stress heterogeneities and force chain anisotropies observed in granular media. Real systems exhibit pronounced directional dependencies in wave velocity and attenuation that MST cannot resolve, particularly under shear-induced anisotropy or uniaxial compression.

Experimental Artifacts and Interpretation Challenges: Experimental applications of MST are compromised by boundary reflections, low signal-to-noise ratios, and oversimplified 2D setups. These artifacts distort coda waveform analyses, leading to unreliable estimates of transport parameters like scattering mean free paths. Furthermore, MST struggles to decouple intrinsic attenuation from scattering effects in polydisperse systems.

Advancements via Nonlinear Elastic Coupling: The current study enhances MST by integrating it with the Jiang–Liu nonlinear elastic theory, enabling simultaneous treatment of ballistic and diffusive wave components. Statistical averaging over multiple disordered configurations reduces microstructural noise, improving robustness in estimating energy diffusivity and scattering parameters. This hybrid approach allows quantitative interpretation of coda waveforms through material-specific transport coefficients.

The proposed framework bridges MST's theoretical foundations with granular-specific complexities, enabling more accurate diagnostics of wave phenomena in realistic particulate media. Future validation against experimental photoelasticimetry and laser Doppler vibrometry data will be critical for refining scattering models.

Effective Medium Theory (EMT)

Effective Medium Theory (EMT) has traditionally provided a foundational framework for predicting elastic wave propagation in granular materials. However, both experimental and computational studies reveal significant shortcomings in its application to real-world systems, particularly under shear or disordered conditions.

Affine Deformation Assumption A core assumption of EMT—*affine deformation*—posits that all grains follow the macroscopic strain field uniformly. While plausible under isotropic compression, this assumption breaks down under shear, where force imbalances induce non-affine displacements such as particle rotations and translations. These localized rearrangements, observed in simulations by Makse et al., are not captured by EMT, leading to inaccuracies in stress and strain predictions.

Static Coordination Number EMT further oversimplifies granular systems by assuming a static, pressure-independent coordination number Z. In reality, Z evolves dynamically with pressure as contacts form or break, following a power-law relationship. This evolution critically influences bulk and shear moduli, yet EMT's static treatment results in underestimation of bulk modulus at high pressures and overestimation of shear modulus at low pressures.

Homogeneity and Isotropy The theory presumes homogeneity and isotropy in contact orientations, ignoring the anisotropic force chains and stress heterogeneities inherent to real granular packings. Such anisotropy generates directional dependencies in wave velocities and stress transmission, phenomena EMT cannot resolve.

Experimental Challenges Experimental validations of EMT face additional challenges. Boundary effects, particle shape variability, frictional heterogeneity, and imperfect contact coupling introduce artifacts in wave measurements. Studies relying on 2D systems or pre-stressed layers further misrepresent the 3D bulk behavior of natural granular media.

Simulation Limitations Even advanced numerical simulations—while more realistic—inherit EMT's limitations. Simplified contact models (e.g., Hertz-Mindlin) neglect critical factors like particle roughness, rolling resistance, and thermal effects. Damping parameters, often calibrated heuristically, fail to replicate energy dissipation mechanisms under dynamic or high-frequency loading.

While EMT offers a valuable first-order approximation, its inability to account for evolving microstructures, non-affine dynamics, and network-level heterogeneity limits its predictive power. These insights underscore the need for advanced frameworks—such as viscoelastic, jamming-based, or nonlocal models—to bridge the gap between idealized theory and the complex reality of granular elasticity.

4.2 Discrete Element Method (DEM)

The two studies by Mouraille et al. (2006) and Mouraille et al. (2008) provide significant insights into sound propagation in granular media using the Discrete Element Method (DEM). However, both investigations exhibit limitations that constrain the generalization of their findings to more realistic and complex granular systems.

Simplified Particle Representations and Contact Models: Both papers assume idealized, spherical particles and implement relatively simplified contact force laws based on linear or piecewise linear spring-dashpot models. Such approximations neglect key nonlinear effects (e.g., Hertzian contact behavior), surface roughness, and rolling resistance. These simplifications limit the accurate reproduction of force chain dynamics and stress distributions that play a crucial role in real granular media.

Static Microstructure Assumptions: The analyses predominantly consider static or quasi-static configurations where the microstructure remains practically unaltered during wave propagation. In contrast, actual granular materials often undergo non-affine deformations and microstructural rearrangements under applied stress or shear. The omission of these dynamic effects—particularly the opening and closing of contacts—reduces the ability of the DEM simulations to capture transient phenomena that can markedly influence both wave speeds and signal attenuation.

Boundary Conditions and Scale Limitations: Both studies utilize simulations on relatively small, highly ordered crystalline packings (e.g., face-centered cubic arrangements of around 3200 particles) with periodic boundary conditions in the transverse directions and fixed layers along the propagation axis. While such idealized geometries facilitate comparison with continuum elasticity theory, they do not fully capture the heterogeneity, anisotropy, and mesoscopic scattering mechanisms present in large-scale or naturally disordered systems.

Influence of Friction, Adhesion, and Polydispersity: Although both papers examine the roles of friction and adhesion, the implementations are rather simplified. For instance, friction is modeled using constant coefficients without considering velocity-dependent friction laws, and the adhesive interactions are represented with elementary elasto-plastic or JKR-type formulations that omit environmental influences (such as humidity). Moreover, the introduction of only slight polydispersity (approximately 0.2% relative to particle size) does not adequately reflect the wider size distributions characteristic of many industrial or geophysical materials. This limited degree of disorder may lead to an overestimated coordination number and insufficient scattering, thereby affecting the simulated wave speed and attenuation.

Stress State and Loading History: In the 2008 study, the evolution of the microstructure during tablet formation and subsequent uniaxial compression is used to assess strain-dependent

acoustic properties. Although variations in wave propagation are observed across different strain regimes, the changes are relatively minor. Such weak strain dependence may be an artifact of the specific preparation protocol and small system size. It remains uncertain whether larger, more heterogeneous systems would display a more pronounced alteration in acoustic behavior as the stress state evolves.

In summary, while DEM simulations have significantly advanced our understanding of sound propagation in granular media, their predictive power is still constrained by idealizations and scale limitations. Addressing non-affine motions, microstructural evolution, and realistic boundary conditions in future models will be vital to developing a more accurate and comprehensive description of acoustic wave propagation in granular systems.

4.3 Force Chain Models

Model Simplifications and Trade-offs The modeling approach in this study involves several trade-offs to balance realism and computational tractability. While angular springs are included to enhance fidelity by capturing additional interparticle interactions, the omission of edge bending simplifies the model at the cost of accuracy in scenarios where bending moments significantly influence stress distribution and wave propagation. This simplification may limit predictive capabilities in systems where flexural rigidity governs force chain evolution.

Neglect of Micropolar Mechanics A critical limitation arises from the exclusion of micropolar mechanics, which account for grain rotations and coupled stresses. This omission overlooks microstructural effects crucial in dense packings, particularly those influencing shear wave modulation and rotational energy transfer. Such effects are known to alter dispersion relations and attenuation profiles in granular media.

Limitations of Effective Medium Theories The evaluation of effective medium theories—including Perfect Bonding (PB), Symmetric Self-Consistent Approximation (SCA-S), and Asymmetric Self-Consistent Approximation (SCA-A)—reveals systematic shortcomings. While bulk modulus predictions remain reasonable, shear moduli estimates fail under high stiffness contrasts or varying volume fractions. For instance, SCA-A underestimates shear stiffness by up to 40% at $\phi = 0.6$ with stiffness ratios $k_2/k_1 > 10^3$, highlighting theoretical gaps in handling heterogeneous contact networks.

Numerical Complexity and Parameter Sensitivity Numerical complexity poses another challenge, particularly in simulations using large periodic Delaunay networks. Balancing computational efficiency with accuracy becomes increasingly difficult when varying parameters like angular stiffness (k_a) or grain stiffness contrast. This complexity obstructs consistent extraction of normalized moduli across volume fractions, complicating universal scaling laws.

Finally, the study remains predominantly theoretical, lacking experimental validation for key predictions such as anomalous Poisson's ratios ($\nu < 0$) in auxetic configurations. Without empirical corroboration, practical implications for sound transmission in real granular systems remain speculative. Future work must integrate experimental acoustics, such as laser Doppler vibrometry or resonant ultrasound spectroscopy, to bridge this validation gap and refine model predictions.

We compare key modeling approaches for acoustic transmission in granular materials, summarizing their capabilities and limitations in Table 4.3.

Table 1: Comparison of Modeling Approaches for Granular Sound Transmission

Feature / Model	**DEM Alone**	**MST**	**Force Chains**	**EMT**	**Hybrid DEM-Darcy-FDM**
Grain-scale behavior	✓	✗	✓	✗	✓
Fluid-particle coupling	✗	✗	✗	✗	✓
Wave propagation	Limited	✓	✗	✓	✓
Nonlinear/Time-varying	✓	✗	Partial	✗	✓
Liquefaction modeling	✗	✗	✗	✗	✓
Computational efficiency	✗	✓	✓	✓	✓
Suitable for large scale	✗	✓	✗	✓	✓

5 Conclusion and Future Work

This study developed a hybrid multiscale modeling framework to investigate wave propagation and attenuation in saturated granular beds by coupling a one-dimensional poromechanical continuum with a grain-scale Discrete Element Method (DEM). Validation against centrifuge experiments involving dry and saturated Nevada Sand showed that the model accurately reproduces wave arrival times, peak accelerations, and pore pressure evolution under seismic loading. Among the many open questions in granular acoustics—such as the effects of non-affine grain motion, saturation heterogeneity, and evolving contact networks—this work specifically addresses how well a coupled DEM–continuum model can simulate stress wave transmission and fluid-solid interactions under transient loading. By solving the momentum and fluid mass balance equations alongside DEM-based contact mechanics, the model captures key aspects of dynamic granular behavior. While results are promising, future work should extend the model to include partial saturation, non-spherical grains, and grain-scale plasticity. Enhanced experimental setups with high-frequency acoustic sensing and real-time pore pressure tracking, as well as data-driven modeling approaches, could further improve predictive capabilities. This work provides a foundation for bridging micro-mechanical insights with macro-scale acoustic response in granular media, offering valuable tools for geophysical and engineering applications. Despite decades of research into wave propagation in granular media, several fundamental challenges remain unresolved—particularly regarding stress wave transmission. Existing methods, whether purely experimental or based on continuum numerical frameworks, often fall short due to their inability to capture microscale dynamics or the nonlinear feedback between solid and fluid phases.

The **Hybrid Multiscale Modeling (HMM)** framework presented in this study provides a transformative approach to addressing these challenges. By coupling a continuum-scale poromechanical solver with grain-resolved Discrete Element Method (DEM) assemblies, this method unifies the strengths of micro- and macro-level analyses into a single, physically consistent, and computationally viable platform.

Why Hybrid DEM–Finite Difference is Superior

As detailed in Kuhn (2021) [17], traditional models suffer from various limitations:

- **Continuum-based FEM models** require complex and sometimes empirically tuned constitutive laws to represent nonlinear behavior, which may not generalize well to new conditions.
- **Multi-physics DEM-CFD models**, while detailed, are computationally expensive and scale poorly for large domains due to the particle-fluid resolution burden.
- **Experimental methods**, such as centrifuge or shake-table tests, though accurate in capturing global trends, cannot provide granular insights into stress redistribution, contact evolution, or pore-level fluid dynamics—especially at the microsecond scale.

In contrast, the hybrid model used here:

- Captures inter-particle dynamics without resolving full Navier–Stokes equations for fluid flow by leveraging Darcy's law for efficient solid-fluid coupling.
- Uses localized DEM representative volume elements (RVEs) instead of filling the entire domain with particles, significantly reducing computational costs while preserving physical realism.
- Includes poromechanically accurate representations of pore fluid pressure evolution, gas solubility, and cavitation phenomena, critical to modeling liquefaction and post-seismic effects.

Issues Resolved by the Hybrid Method

This hybrid method addresses numerous limitations of existing approaches:

- **Energy Dissipation and Plasticity:** Allows energy losses to arise naturally through contact dynamics, damping, and plastic slip, rather than from assumed damping factors.
- **Nonlinearity from First Principles:** Stress-strain nonlinearities emerge from contact mechanics and fluid coupling, avoiding reliance on empirical constitutive laws.
- **Multiphase Effects:** Accurately captures the role of entrapped air, pore collapse, and quasi-saturation states in shifting wave behavior and liquefaction thresholds.
- **Wave-Speed Modulation:** Demonstrates how softening due to increasing pore pressures reduces wave velocities and alters frequency content—key experimental observations.

By uniting DEM-based granular mechanics with continuum-scale wave and pressure solvers, the hybrid multiscale model not only overcomes existing methodological bottlenecks but also opens new avenues for understanding and predicting complex soil-fluid interactions under dynamic loading. It represents a robust and extensible framework for future studies in earthquake geotechnics, wave mechanics, and granular material science.

6 Appendix: Notation and Symbol Definitions

6.1 Glossary of Symbols

Symbol	Definition
ρ	Bulk density of the soil-solid-fluid mixture [kg/m³]
u	Displacement of the solid phase [m]
t	Time [s]
σ	Total stress in the solid matrix [Pa]
z	Vertical spatial coordinate [m]
g	Acceleration due to gravity [m/s²]
p	Pore fluid pressure [Pa]
n	Porosity of the soil (volume fraction of voids) [-]
β_w	Compressibility of the pore fluid [Pa^{-1}]
α	Biot coefficient (fluid-solid coupling term) [-]
k	Permeability of the porous medium [m²]
M	Biot's modulus (bulk modulus of the porous medium) [Pa]
ε_v	Volumetric strain (from DEM) [-]
m_i	Mass of particle i in DEM model [kg]
x_i	Position of particle i [m]
f_{ij}^c	Elastic contact force between particles i and j [N]
f_{ij}^d	Damping (viscous) force between particles i and j [N]
δ	Normal overlap or deformation between particles [m]
k_n	Normal stiffness constant for contact [N/m$^{3/2}$]
η	Damping coefficient [Ns/m]
σ'	Effective stress (carried by the soil skeleton) [Pa]
Δt	Time step in numerical scheme [s]
Δz	Spatial discretization step [m]
u^n	Displacement at time step n [m]
p^n	Pore pressure at time step n [Pa]
k_n	Normal (central) spring stiffness [N/m]
k_a	Angular spring stiffness [Nm/rad^2]
U_{central}	Energy stored in central springs [J]
U_{angular}	Energy stored in angular springs [J]
U	Total strain energy in the system [J]

$\mathbf{u}_i$	Displacement vector of grain i [m]
$\hat{n}_{ij}$	Unit vector from grain i to j [-]
$\Delta\phi$	Angular change between adjacent edges [rad]
$G(V, E)$	Delaunay graph with vertices V and edges E
K	Effective bulk modulus of the granular medium [Pa]
μ	Effective shear modulus of the granular medium [Pa]
E	Effective planar Young's modulus [Pa]
ν	Effective Poisson's ratio [-]
c_1, c_2	Volume fractions of soft and stiff grains [-]
r	Stiffness contrast: k_{n2}/k_{n1} [-]
sr	Dimensionless stiffness ratio: $\dfrac{k_n l^2}{k_a}$ [-]
l	Characteristic edge length in Delaunay network [m]
κ	Effective bulk modulus in effective medium theory [Pa]
ϕ	Angular coordinate or grain rotation angle [rad]
ε	Macroscopic strain tensor [-]
$\Delta u(e)$	Relative displacement along edge e [m]
Δt	Time step in numerical simulation [s]
δ	Normal overlap between two contacting particles
δ_{max}	Maximum historical overlap during contact
δ_*	Overlap at which the force becomes zero for current stiffness
δ_f	Overlap at which the force vanishes when using stiffness k_2
a_i, a_j	Radii of particles i and j
$\mathbf{r}_i, \mathbf{r}_j$	Position vectors of particles i and j
$\mathbf{n}$	Unit normal vector at the point of contact
f_n	Normal contact force
f_t	Tangential contact force
k_1	Initial loading stiffness
k_2	Maximum elastic stiffness
k_c	Adhesive stiffness constant
k_t	Tangential stiffness constant
γ_n	Normal damping coefficient
γ_t	Tangential damping coefficient
μ	Friction coefficient
$\mathbf{v}_i, \mathbf{v}_j$	Linear velocities of particles i and j

$\boldsymbol{\omega}_i, \boldsymbol{\omega}_j$	Angular velocities of particles i and j
$\mathbf{v}_{ij}$	Relative velocity at the contact point
$\mathbf{v}_t$	Tangential component of the relative velocity
$\boldsymbol{\xi}$	Tangential displacement accumulated over time
c_b	Background damping coefficient (translation)
c_{br}	Background damping coefficient (rotation)
$\mathbf{f}_i$	Total force on particle i
$\boldsymbol{\tau}_i$	Torque on particle i
m	Mass of a particle
Δt	Time step size
C	Safety factor for time integration stability
v_p	Compressional (P-wave) velocity [m/s]
v_s	Shear (S-wave) velocity [m/s]
K	Bulk modulus of the granular medium [Pa]
μ	Shear modulus of the granular medium [Pa]
ρ	Bulk density of the medium [kg/m³]
m_i	Mass of particle i [kg]
$\vec{r}_i$	Position vector of particle i [m]
$\vec{F}_{ij}^n$	Normal contact force between particles i and j [N]
$\vec{F}_{ij}^t$	Tangential contact force between particles i and j [N]
$\vec{F}_{ij}^d$	Damping (viscous) force between particles i and j [N]
F_n	Normal contact force (Hertzian model) [N]
F_t	Tangential force (Mindlin model) [N]
k_n	Normal stiffness constant [N/m$^{3/2}$]
k_t	Tangential stiffness constant [N/m$^{1/2}$]
R	Effective radius of contacting spheres [m]
δ	Normal overlap (deformation) [m]
s	Tangential displacement or slip distance [m]
a	Contact radius between particles [m]
E^*	Effective Young's modulus [Pa]
ν	Poisson's ratio [-]
Z	Coordination number (average number of contacts per particle) [-]
$\vec{u}$	Displacement vector of the solid phase [m]
$\vec{U}$	Displacement vector of the fluid phase [m]

σ	Stress tensor in the solid phase [Pa]
p	Pore pressure [Pa]
ρ_{11}, ρ_{12}, ρ_{22}	Coupled density coefficients in Biot's theory [kg/m³]
α	Biot coefficient [-]
b	Viscous drag coefficient between solid and fluid [Ns/m^4]
K_{sat}	Bulk modulus of saturated medium [Pa]
K_{dry}	Bulk modulus of dry granular skeleton [Pa]
K_s	Bulk modulus of the solid grains [Pa]
K_f	Bulk modulus of the pore fluid [Pa]
ϕ	Porosity (volume fraction of voids) [-]
$\rho(\vec{r})$	Mass density at position $\vec{r}$ [kg/m³]
u_i	Displacement component in the i-th direction [m]
t	Time [s]
σ_{ij}	Stress tensor component [Pa]
$f_i(\vec{r}, t)$	External body force acting at $\vec{r}$ [N]
$\lambda(\vec{r})$	First Lamé parameter at $\vec{r}$ [Pa]
$\mu(\vec{r})$	Shear modulus (second Lamé parameter) at $\vec{r}$ [Pa]
δ_{ij}	Kronecker delta (identity tensor component) [-]
ϵ_{ij}	Strain tensor component [-]
ϵ_{ll}	Volumetric strain (trace of strain tensor) [-]
ϵ^0_{ij}	Deviatoric part of the strain tensor [-]
ϵ^2_s	Second invariant of deviatoric strain [-]
K	Bulk modulus [Pa]
G	Shear modulus [Pa]
$\tilde{K}, \tilde{G}$	Moduli constants in Jiang–Liu model [Pa]
a, b	Nonlinearity exponents in Jiang–Liu model [-]
δ	Compression: $\delta = -\epsilon_{ll}$ [-]
$F(\epsilon)$	Free energy density [J/m³]
$G(\omega)$	Green's function at frequency ω [-]
ω	Angular frequency [rad/s]
ϵ	Small positive number ensuring convergence [-]
$\mathcal{K}$	Wave operator (kernel) [-]
$G_0(\omega)$	Green's function for homogeneous medium [-]
$\Sigma(\omega)$	Self-energy representing scattering effects [-]

$\delta\mathcal{K}$	Fluctuation in wave operator [-]
L	Intensity correlation function (Green's function second moment) [-]
U	Irreducible vertex function (scattering interactions) [-]
τ_j	Scattering time for mode j [s]
$L_j(\vec{x}, t)$	Intensity at location $\vec{x}$ and time t for mode j [-]
S_j	Source strength for mode j [-]
$W_{jj'}$	Scattering kernel from mode j' to j [-]
I	Wave intensity field [-]
D	Diffusion coefficient [m²/s]
τ_a	Attenuation time [s]
Q^{-1}	Inverse quality factor (dissipation measure) [-]
ℓ	Scattering length [m]
k	Wavenumber [1/m]
d	Particle diameter [m]

References

[1] Khaled I Alzebdeh, Nabil G Chalhoub, and C Darwin Foster. A spring network model for elastic properties of granular materials. *Mechanics of Materials*, 31(6):375–387, 1999.

[2] K. Arulmoli, K. Muraleetharan, M. Hossain, and R. Fragaszy. Velacs: Verification of liquefaction analyses by centrifuge studies—laboratory testing program. Technical report, University of California, Davis, 1992.

[3] RJ Bathurst and L Rothenburg. Micromechanical aspects of isotropic granular assemblies with friction. *Journal of Applied Mechanics*, 55(1):17–23, 1988.

[4] M. A. Biot. General theory of three-dimensional consolidation. *Journal of Applied Physics*, 12(2):155–164, 1941.

[5] MA Biot. Theory of propagation of elastic waves in a fluid-saturated porous solid. i. low-frequency range. *The Journal of the acoustical Society of America*, 28(2):168–178, 1956.

[6] MA Biot. Mechanics of deformation and acoustic propagation in porous media. *Journal of Applied Physics*, 33(4):1482–1498, 1962.

[7] N. V. Brilliantov, F. Spahn, J. M. Hertzsch, and T. Pöschel. Model for collisions in granular gases. *Physical Review E*, 53(5):5382, 1996.

[8] Bernard Budiansky. On the elastic moduli of some heterogeneous materials. *Journal of the Mechanics and Physics of Solids*, 13(4):223–227, 1965.

[9] P. A. Cundall and O. D. L. Strack. A discrete numerical model for granular assemblies. *Geotechnique*, 29(1):47–65, 1979.

[10] M. Fehler and H. Sato. Seismic wave propagation in heterogeneous crust: A review. *Geophysical Monograph-American Geophysical Union*, 157:3–18, 2003.

[11] Fritz Gassmann. Elastic waves through a packing of spheres. *Vierteljahrsschrift der Naturforschenden Gesellschaft in Zürich*, 96:1–23, 1951.

[12] Rodney Hill. A self-consistent mechanics of composite materials. *Journal of the Mechanics and Physics of Solids*, 13(4):213–222, 1965.

[13] J. C. Jaeger and K. L. Johnson. *Contact mechanics*. Cambridge University Press, 1996.

[14] Anand Jagota and Chung-Yuen Hui. Spring network models of the elastic constants of disordered solids. *International Journal of Solids and Structures*, 31(3):335–349, 1994.

[15] Xiaoping Jia, Christian Caroli, and B Velicky. Ultrasound propagation in granular media: Scattering, absorption and diffusion. *Physical Review Letters*, 82(9):1863–1866, 2009.

[16] KL Johnson. Contact mechanics. *Cambridge university press*, 1985.

[17] Matthew R. Kuhn. Rate dependence and numerical efficiency in quasi-static dem simulations. *Computers and Geotechnics*, 135:104168, 2021.

[18] Matthew R. Kuhn. Multi-scale simulation of wave propagation and liquefaction in a one-dimensional soil column: hybrid dem and finite-difference procedure. *Acta Geotechnica*, 17(6):2611–2632, 2022.

[19] H. Kwon. *Cyclic simple shear behavior of Nevada Sand and verification of a constitutive model*. PhD thesis, University of California, Davis, 2008.

[20] S. Luding. Cohesive, frictional powders: contact models for tension. *Granular Matter*, 10:235–246, 2008.

[21] Hernán A Makse, N Gland, David L Johnson, and Lawrence M Schwartz. Stress-dependent elastic moduli in granular packings. *Physical Review Letters*, 83(24):5070, 1999.

[22] O. Mouraille, O. Herbst, and S. Luding. Sound propagation in isotropically and uniaxially compressed cohesive, frictional granular solids. *Engineering Fracture Mechanics*, 76(5):781–792, 2009.

[23] O. Mouraille, W. A. Mulder, and S. Luding. Sound wave acceleration in granular materials. *Journal of Statistical Mechanics: Theory and Experiment*, 2006(07):P07023, 2006.

[24] Martin Ostoja-Starzewski. Linear elastic behavior of random networks: New results. *Mechanics of Materials*, 20(2):129–143, 1995.

[25] Lenya Ryzhik, George Papanicolaou, and Joseph B Keller. Transport equations for elastic and other waves in random media. *Wave Motion*, 24(4):327–370, 1996.

[26] M F Thorpe, P M Duxbury, and D J Jacobs. Elastic properties of bidisperse random networks. *Journal of Non-Crystalline Solids*, 75(1-3):841–848, 1985.

[27] Leonardo Trujillo, María Peniche, and Leonardo Di G Sigalotti. Green's function approach to the propagation of acoustic waves in granular media. *Physica A: Statistical Mechanics and its Applications*, 389(24):5353–5366, 2010.

[28] Y. Tsuji, T. Tanaka, and T. Ishida. Lagrangian numerical simulation of plug flow of cohesionless particles in a horizontal pipe. *Powder Technology*, 71:239–250, 1992.

[29] Vincenzo Vitelli, Ning Xu, Matthieu Wyart, Andrea J Liu, and Sidney R Nagel. Topological mechanisms in jammed matter. *Physical Review E*, 81(2):021301, 2010.

Author Contributions

The authors contributed to this work through collaboration across the major sections of the paper. The breakdown of contributions is as follows:

- **Introduction:** Moni and Jasraj were responsible for writing the introduction, establishing the context for the study and outlining the motivation behind exploring sound transmission in granular media.
- **Literature Review:** Vyom and Shrimi conducted an in-depth review of the relevant literature, identifying key theoretical and experimental advancements in the field, and highlighting existing gaps that the current study addresses.
- **Methods:** All authors contributed equally to the development and documentation of the methodological framework, including the application of effective medium theory (EMT), multiple scattering theory (MST), force chain analysis, and discrete element simulations.
- **Discussion:** The discussion of results, including interpretation of simulations and theoretical insights, was a collective effort by the entire team, ensuring a comprehensive understanding of the study's implications.
- **Conclusion:** Jay and Nikhil prepared the conclusion, summarizing the key findings, outlining limitations, and suggesting future directions for research on sound propagation in granular materials.